HDOC Day-60

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Problem Statement

Write a C program that reads two strings, **s** and **t**. Assume both strings contain only lowercase characters. Pass the strings to a function named **check_anagram()** using their addresses. Use pointers to pass the strings between the calling function and **check_anagram()**.

The function must determine whether both strings contain the same characters with the same frequencies. The order of characters is irrelevant. Print **Anagram** for valid anagrams and **Not Anagram** otherwise. Strings of different lengths must be treated as not anagrams. Use a character-frequency-counting approach.

C Program

```
#include <stdio.h>
#include <string.h>

void check_anagram(char *s, char *t)
{
    int freq[26] = {0};
    int i;

    if (strlen(s) != strlen(t))
    {
        printf("Not Anagram\n");
        return;
    }

    for (i = 0; *(s + i) != '\0'; i++)
    {
        freq[*(s + i) - 'a']++;
        freq[*(t + i) - 'a']--;
    }

    for (i = 0; i < 26; i++)
    {
        if (freq[i] != 0)
        {
            printf("Not Anagram\n");
            return;
        }
    }

    printf("Anagram\n");
}

int main()
{
    char s[1000], t[1000];

    printf("Enter string s: ");
    scanf("%999s", s);

    printf("Enter string t: ");
    scanf("%999s", t);

    check_anagram(s, t);

    return 0;
}
```

Explanation

The addresses of strings **s** and **t** are passed to **check_anagram()** as pointers. First, their lengths are compared. A frequency array of 26 positions is then used for the lowercase English letters. Each character from **s** increases its frequency while the corresponding character from **t** decreases it. If every final frequency is zero, both strings contain exactly the same characters the same number of times and are anagrams.

Sample Outputs

```
Example 1
String s: listen
String t: silent
Output: Anagram

Example 2
String s: hello
String t: world
Output: Not Anagram

Example 3
String s: anagram
String t: nagaram
Output: Anagram

Example 4
String s: rat
String t: car
Output: Not Anagram
```

Result

Thus, the program successfully checks whether two lowercase strings are anagrams using pointers and character-frequency counting.